

# PABLO M. BERNÁ

PHD IN MATHEMATICS · ASSOCIATE PROFESSOR

pablomanuel.berna@utamed.es · pablomates.github.io · ORCID



## SUMMARY

PhD in Mathematics with International Mention (UAM), specialized in Functional Analysis and Approximation Theory. My research focuses on greedy algorithms in Banach and quasi-Banach spaces, with more than 30 publications in high-impact journals. I currently combine my position as Associate Professor at UTAMED University with institutional leadership roles as Director of the Master's Degree in Artificial Intelligence and Director of the Teaching and Research Staff (PDI) Department within the Vice-Rectorate for Academic Affairs, as well as at the Spanish Royal Mathematical Society (RSME). My commitment to academic excellence is reflected in the supervision of doctoral theses and continuous participation in competitive national research projects.

## EDUCATION

- PhD in Mathematics (Cum Laude, International Mention)** 2017–2020  
Universidad Autónoma de Madrid. *"Thresholding greedy algorithms in Banach spaces"*.
- Master's Degree in Education: Mathematics and Computer Science** 2018–2019  
International University of Valencia.
- Master's Degree in Mathematical Research** 2014–2015  
University of Valencia.
- Bachelor's Degree in Mathematics** 2009–2014  
University of Alicante.

## MANAGEMENT & ACCREDITATIONS

- Director of the Master's Degree in Artificial Intelligence** 2026–Present  
UTAMED University.
- Director of the Teaching and Research Staff (PDI) Department** 2026–Present  
UTAMED University, Vice-Rectorate for Academic Affairs.
- Deputy Secretary** 2025–Present  
Spanish Royal Mathematical Society (RSME).
- Deputy Director of the Polytechnic School** 2024–2026  
CUNEF University.
- Editorial Boards**  
*Demonstratio Mathematica* (Q1, JCR) and *Journal of Mathematical Analysis*.
- ANECA Accreditations**  
Associate Professor (Profesor Titular) and Tenured Contracted Professor (Profesor Contratado Doctor).
- Research Evaluation Period (Sexenio)**  
Recognized six-year research period 2018–2023.

## SCIENTIFIC PUBLICATIONS

35. F. Albiac, J. L. Ansorena, M. Berasategui, **P. M. Berná**. *When Greedy Approximation Breaks: Counterexamples in Quasi-Banach Spaces*. Accepted in Transactions of the American Mathematical Society (2026).
34. M. Berasategui, **P. M. Berná**, H. V. Chu, A. García. *Lebesgue-type estimates for greedy algorithms in quasi-Banach spaces*. Rev Mat Complut (2026). <https://doi.org/10.1007/s13163-026-00587-3>.
33. F. Albiac, J. L. Ansorena, M. Berasategui, **P. M. Berná**. *Conditional bases with Property (A)*. Accepted in Constructive Approximation (March 2026).
32. **P. M. Berná**. *Step-Size Decay and Structural Stagnation in Greedy Sparse Learning*. Mathematics 2026, 14(6), 967.
31. **P. M. Berná**, A. García. *The Weak TGA in  $p$ -Banach spaces*. Dolomites Research Notes on Approximation, 19(1), 48–55 (2026).
30. M. Berasategui, **P. M. Berná**, H. V. Chu. *On sequential greedy-like bases*. Ann. Funct. Anal. 16, 36 (2025).
29. **P. M. Berná**, H. V. Chu, E. Hernández. *On approximation spaces and Greedy-type bases*. Ann. Funct. Anal. (2025) 16:7.
28. M. Berasategui, **P. M. Berná**, D. González. *Approximation by polynomials with constant coefficients and the Thresholding Greedy Algorithm*. Carpathian J. Math. 41, no. 1 (2025).
27. **P. M. Berná**, A. Falcó. *Universality for non-linear convex variational problems*. J. Nonlinear Var. Anal. 9 (2025), 161–177.
26. M. Berasategui, **P. M. Berná**, H. V. Chu. *On consecutive greedy and other greedy-like type of bases*. Res. Math. Sci. 11, 61 (2024).
25. **P. M. Berná**, A. García, D. González. *A note on 1-semi-greedy bases in  $p$ -Banach spaces with  $0 < p \leq 1$* . Demonstratio Mathematica, 57(1), 2024, 20240047.
24. M. Berasategui, **P. M. Berná**. *Greedy-like bases for sequences with gaps*. Banach J. Math. Anal. 18, 14 (2024).
23. **P. M. Berná**, D. González. *Non-linear approximation by 1-greedy bases*. J. Math. Anal. Appl. 531(1), (2024), 127769.
22. M. Berasategui, **P. M. Berná**, H. V. Chu. *Extensions and new characterizations of some greedy-type bases*. Bull. Malay. Math. Sci. Soc. 46, 84 (2023).
21. F. Albiac, J. L. Ansorena, M. Berasategui, **P. M. Berná**, S. Lassalle. *Bidemocratic bases and their connections with other greedy-type bases*. Constr. Approx. 57 (2023), 125–160.
20. F. Albiac, J. L. Ansorena, **P. M. Berná**. *New parameters and Lebesgue-type estimates in greedy approximation*. Forum of Mathematics Sigma 10 (2022), e113, 1–39.
19. M. Berasategui, **P. M. Berná**. *Extensions of democracy-like properties for sequences with gaps*. Math. Inequal. Appl. 25(4) (2022), 1155–1189.
18. **P. M. Berná**, H. V. Chu. *On some characterization of greedy-type bases*. Expo. Math. 40 (2022), 1135–1158.
17. F. Albiac, J. L. Ansorena, M. Berasategui, **P. M. Berná**, S. Lassalle. *Weaker forms of unconditionality of bases in greedy approximation*. Studia Math. 257(1) (2022), 1–17.
16. **P. M. Berná**, D. Mondéjar. *A functional characterization of almost-greedy bases*. Mathematics (2021) 9(15):1827.
15. M. Berasategui, **P. M. Berná**. *Quasi-greedy bases for sequences with gaps*. Nonlinear Analysis 208 (2021), 112294.
14. F. Albiac, J. L. Ansorena, **P. M. Berná**, P. Wojtaszczyk. *Greedy approximation for biorthogonal systems in quasi-Banach spaces*. Dissert. Math. 560 (2021), 1–88.
13. **P. M. Berná**. *A note on partially-greedy bases in quasi-Banach spaces*. Studia Math. 259 (2021), 225–239.
12. M. Berasategui, **P. M. Berná**, S. Lassalle. *Strong-partially-greedy bases and Lebesgue-type inequalities*. Constr. Approx. 54 (2021), 507–528.
11. **P. M. Berná**, J. D. Rossi. *Nonlocal diffusion equations with dynamical boundary conditions*. Nonlinear

Analysis 195 (2020), 111751.

10. F. Albiac, J. L. Ansorena, **P. M. Berná**. *Asymptotic greediness of the Haar System in the  $L_p$  spaces for  $1 < p < \infty$* . Constr. Approx. 51, 427–440 (2020).
9. **P. M. Berná**. *Characterization of weight-semi-greedy bases*. J. Fourier Anal. Appl. 26, 21 (2020).
8. **P. M. Berná**, Ó. Blasco, G. Garrigós, E. Hernández, T. Oikhberg. *Lebesgue inequalities for Chebyshev Thresholding Greedy Algorithms*. Rev. Matem. Complut. 33, 695–722 (2020).
7. **P. M. Berná**, D. Kutzarova, S. J. Dilworth, T. Oikhberg, B. Wallis. *The weighted Property (A) and the greedy algorithm*. J. Approx. Theory 248 (2019), 105300.
6. **P. M. Berná**, A. Pérez. *A remark on approximation with polynomials and greedy bases*. J. Math. Anal. Appl. 478(2) (2019), 466–475.
5. **P. M. Berná**. *Equivalence between almost-greedy and semi-greedy bases*. J. Math. Anal. Appl. 417 (2019), 218–225.
4. **P. M. Berná**, Ó. Blasco, G. Garrigós, E. Hernández, T. Oikhberg. *Embeddings and Lebesgue inequalities for greedy algorithms*. Constr. Approx. 48(3) (2018), 415–451.
3. **P. M. Berná**, Ó. Blasco, G. Garrigós. *Lebesgue inequalities for the greedy algorithm in general bases*. Rev. Mat. Complut. 30 (2017), 369–392.
2. **P. M. Berná**, Ó. Blasco. *The best  $m$ -term approximation with respect to polynomials with constant coefficients*. Anal. Math. 43(2) (2017), 119–132.
1. **P. M. Berná**, Ó. Blasco. *Characterization of greedy bases in Banach spaces*. J. Approx. Theory 215 (2017), 28–39.

## SUBMITTED PAPERS

---

1. **P. M. Berná**. *Greedy spectral-coordinate Frank-Wolfe over the Spectrahedron* (submitted 2026).
2. **P. M. Berná**, A. Falcó, D. Mondéjar. *Resolution-Consistent Greedy Neural Approximation on Infinite-Dimensional Spaces* (submitted 2026).
3. M. Berasategui, **P. M. Berná**, D. Kutzarova, T. Oikhberg. *Characterizations of the unit vector basis of  $c_0$  or  $\ell_p$  among minimal systems* (submitted 2026).
4. **P. M. Berná**, A. Falcó, D. Mondéjar. *Anytime-Valid Residual Certification for Adaptive Randomized Low-Rank Approximation* (submitted 2026).
5. **P. M. Berná**, A. Tallab, K. Hamidi, A. Mazouz. *Lipschitz adjoints and quasi-compact operators in  $p$ -Banach spaces* (submitted 2026).
6. **P. M. Berná**. *Geometry-Weighted Orthogonal Greedy Approximation on Sampled Manifolds: Stability, Consistency, and Numerical Experiments* (submitted 2026).
7. **P. M. Berná**, A. García. *Noise-Aware Stopping for the Weak Rescaled Pure Greedy Algorithm: Discrepancy Rules and Support Preservation under Incoherence* (submitted 2026).
8. F. Albiac, J. L. Ansorena, M. Berasategui, **P. M. Berná**. *Isometric renormings for greedy bases in Banach spaces, with applications to the Haar System in  $L_p[0, 1]$ ,  $1 < p < \infty$* .
9. **P. M. Berná**, A. García. *Convergence Analysis of Greedy Algorithms with Adaptive Relaxation in Hilbert Spaces* (submitted 2026).
10. **P. M. Berná**, D. Freeman, T. Oikhberg, M. A. Taylor. *Maximal inequalities, frames and greedy algorithms* (submitted 2026).
11. M. Berasategui, **P. M. Berná**, A. García. *Approximation spaces, greedy classes and Lorentz spaces* (submitted 2025).
12. M. Berasategui, **P. M. Berná**, S. J. Dilworth, D. Kutzarova. *Summability Methods for the Greedy Algorithm in Banach spaces* (submitted 2025).

## WORKS IN PROGRESS

---

1. M. Berasategui, **P. M. Berná**, A. García. *Lebesgue-type estimates for the WXCGA*.

2. **P. M. Berná**, A. García, D. Mondéjar. *Some results about 1-semi-greedy bases*.
3. **P. M. Berná**, A. García, D. Mondéjar. *Greedy algorithms in the context of TDA*.
4. **P. M. Berná**, A. García. *Weighted greedy bases in  $p$ -Banach spaces*.
5. **P. M. Berná**, A. García. *Review of greedy algorithms in Hilbert spaces*.
6. **P. M. Berná**, A. García. *Step-size decay in greedy sparse learning*.

## PHD SUPERVISION

- **Andrea García Pons** Ongoing, 2024–2027  
*Greedy Algorithms and approximation spaces*. CEU Universities. Co-direction with Miguel Berasategui.
- **David González Moro** Completed, 2023–2026  
*Mathematical Foundations of Greedy Algorithms in quasi-Banach spaces*. CEU Universities. Sobre-saliente Cum Laude.
- **Hung Viet Chu** Completed, 2019–2023  
*Relaxations of the optimality requirement on the Thresholding Greedy Algorithm for bases of Banach spaces*. University of Illinois at Urbana-Champaign (Co-advisor).

## RESEARCH VISITS

- **University of Illinois at Urbana-Champaign, USA** 2017 & 2023 & 2025  
 September–October 2017, March 2023, April 2025.
- **Isaac Newton Institute, Cambridge, UK** 2024  
 July 2024.
- **University of Cambridge, UK** 2019  
 February–May 2019.
- **University of Buenos Aires, Argentina** 2019  
 August–October 2019.

## RESEARCH PROJECTS

- **Análisis de Fourier y Aplicaciones**  
 Agencia Estatal de Investigación (AEI). Ref. PID2022-142202NB-100.
- **Análisis de Fourier y Aplicaciones**  
 Agencia Estatal de Investigación (AEI). Ref. PID2019-105599GB-100.
- **La interacción entre geometría y topología en espacios de Banach y sus aplicaciones**  
 MINECO. Ref. MTM2017-83262-C2-2-P.
- **Interacción y aplicaciones en Análisis Funcional y Armónico**  
 Fundación Séneca. Ref. 20906/PI/18.
- **Análisis de Fourier, Análisis Real y Aplicaciones**  
 Ref. MTM2016-76566-P.
- **Interacción y aplicaciones en Análisis Funcional y Armónico**  
 Fundación Séneca. Ref. 19368/PI/14.
- **Exploria Teaching Innovation Project**  
 Innovation in the teaching of Mathematics in Economics and Business degrees. CEU San Pablo University (2020–2021).

## INVITED TALKS & CONFERENCES

---

I have delivered nearly 50 contributions at national and international conferences and seminars.

### SELECTION OF INVITED TALKS

- **Renormings and greedy bases** Sep 2026  
Thirteenth Conference on Mathematics and its Applications, Puebla (México).
- **Universality for non-linear convex variational problems** Feb 2025  
Congreso de Jóvenes Investigadores de la RSME 2025, Bilbao, Spain.
- **Recent advances about greedy-like bases** Jul 2023  
Sub-Riemannian Geometry, Harmonic Analysis, PDEs and Applications, Università di Bologna, Italy.
- **The bideocracy in the world of greedy approximation** Sep 2021  
Approximation and Discretization (Online), Moscow.
- **Lebesgue-type estimates for the Thresholding Greedy Algorithm** Nov 2021  
New Bridges between Mathematics and Data Science, Valladolid.
- **La avaricia de la base de Haar** Jun 2019  
EARCO 2019, Universidad de Valencia.

### SOME CONTRIBUTIONS (TALKS, POSTERS AND SEMINARS)

- **Approximation spaces and the greedy algorithm** Jul 2026  
FoCM 2026, Vienna.
- **Introduction to the TGA in Banach spaces** Apr 2025  
Learning Seminar in Banach spaces, University of Illinois at Urbana-Champaign, USA.
- **Greedy-like bases for sequences with gaps** Jan 2024  
Congreso Bienal de la RSME 2024, Pamplona, Spain.
- **Approximation spaces and greedy-like bases** Jun 2023  
Foundations of Computational Mathematics, Paris.
- **New advances in greedy approximation theory** Apr 2023  
Analysis Seminar, University of Illinois at Urbana-Champaign, USA.
- **New results about Lebesgue-type parameters** Oct 2022  
International E-Conference: A Selçuk Meeting, Selçuk University (Online).
- **La condición de la avaricia en aproximación** Sep 2019  
SUMA 2019, Mendoza, Argentina.

### ORGANIZATION OF SESSIONS AND CONFERENCES

- **Parallel Session, RSME Young Researchers Conference 2025** Feb 2025  
Numerical Methods for Applied Mathematics, University of Bilbao, Spain.
- **Parallel Session, RSME Biennial Congress 2024** Jan 2024  
Numerical Methods and Computational Mathematics, Pamplona, Spain.
- **Applied Mathematics and Statistics Seminar** 2020–2021  
Member of the organizing committee, CEU San Pablo University.

### SCIENTIFIC REVIEWING

---

Regular referee for: Mathematical Inequalities and Applications, Journal of Mathematical Analysis and Applications, Demonstratio Mathematica, Revista Matemática Complutense, Mathematics, Advances in Operator Theory, Zeitschrift für angewandte Mathematik und Physik, Annals of Functional Analysis, Jaen Journal on Approximation and Mathematical Reviews (AMS).

### TEACHING

---

I have taught more than 2000 hours (Spanish and English) at different universities and degree programs.

- **UTAMED** 2025–Present  
Mathematics for AI and Deep Learning (Master in Artificial Intelligence) and different mathematical subjects in several degrees of the university.
- **CUNEF University** 2022–2026  
Mathematical Analysis II and III (Mathematical Engineering), Calculus II (Data Science), Statistical Inference (Mathematical Engineering), Data Analysis and Probability and Statistical Inference (Computer Engineering).
- **CEU San Pablo University** 2020–2022  
Mathematics I and II (Business Administration), Dynamical Systems and Optimization Theory (Economics).
- **European University** 2020–2022  
Calculus II (Aerospace), Statistics (Physics), Biostatistics (Biomedical Engineering) and Mathematics I (Business Administration).

Additionally, I have supervised several undergraduate theses on the interaction between Mathematics and Economics, as well as a Master's thesis on greedy algorithms.